

Mihir Ravindra Adelkar

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Summary

Software Engineer with 3+ years building scalable AI systems integrating LLMs. Expert in Python, FastAPI, RAG patterns, and MCP Architecture. Built a production Prior Authorization AI system, reducing processing from 3-5 days to 5 seconds with 95%+ accuracy. Strong full-stack foundation (React/TypeScript, Node.js) with proven ability to architect scalable microservices and deliver solutions.

Skills

AI & LLMs	RAG, MCP, Claude API, LLM Integration (Llama 3), Prompt Engineering, Vector DBs (Pinecone), Embeddings, NLP
Backend	Python (FastAPI/Django/Flask), Node.js (Express/NestJS), Java (Spring Boot), Go, RESTful APIs, GraphQL, gRPC
Frontend	React, Next.js, TypeScript, Redux, Material-UI, Tailwind CSS, Responsive Design
Data	PostgreSQL, MySQL, MongoDB, Redis, Schema Design, Query Optimization
DevOps	AWS (EKS, Lambda, S3, RDS), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD
Tools	Jest, Cypress, TDD, K6 Load Testing, Prometheus, Grafana, New Relic, SonarCloud

Professional Experience

Full-Stack Developer , KeelWorks Foundation — Boston, MA	Aug 2025 – Present
- Developing workforce engagement platform using React/TypeScript frontend and Node.js microservices with PostgreSQL. - Leveraging AI assistants to accelerate development; building custom AI agents for code scaffolding and automation. - Deploying containerized services on AWS EKS with Docker and Terraform; implementing CI/CD pipelines with GitHub Actions.	
Software Engineer , Freelancing (Dabadu.ai) — Remote	May 2023 – Sept 2023
- Contributed to automotive dealership CRM with React/Node.js, implementing real-time features using WebSockets. - Integrated Elasticsearch for full-text search; refactored JavaScript to TypeScript for improved maintainability. - Established Cypress E2E testing and SonarCloud quality gates in CI/CD pipeline, reducing production bugs by 20%.	
Software Engineer , NeoSOFT Technologies — Mumbai, India	Jun 2021 – May 2023
- Architected full-stack applications using React/TypeScript, Node.js, Django, and Spring Boot across CRM and analytics platforms serving 500K+ users. - Designed RESTful and GraphQL APIs with PostgreSQL/MongoDB; implemented Redis caching reducing response times by 70%. - Led migration from monolithic architecture to Docker containerized microservices, reducing deployment time by 80%. - Implemented Kafka for async processing (1B+ messages/day) and gRPC for low-latency inter-service communication. - Built reusable component libraries published as internal npm packages, adopted by 15+ teams for accelerated development. - Established CI/CD pipelines with Jenkins and GitHub Actions; achieved 85% test coverage with Jest/Cypress automation. - Added observability with Prometheus/Grafana and New Relic; reduced mean-time-to-recovery from 25 to 15 minutes. Mentored 8 junior developers through code reviews, pair programming, and technical guidance in Agile sprints.	

Projects

Healthcare Prior Authorization AI System	Python, FastAPI, Claude API, FHIR R4, MCP, Streamlit, Tesseract OCR
- Engineered AI system transforming 3-5 day manual healthcare prior authorization reviews into automated 5-second decisions, processing 100+ requests per day and reducing manual review workload by 80%. - Architected clinical NLP pipeline with FHIR R4-compliant resource mapping, extracting structured medical data (CPT codes, medications, diagnoses, contraindications) from unstructured clinical notes with 95%+ accuracy. - Designed Model Context Protocol (MCP) architecture for intelligent context management, combining patient clinical data with insurance coverage guidelines for LLM-based decision making with full audit trails, citation traceability, and PHI handling.	
CVE Intelligence System — RAG-Powered Security Assistant	Go, Python, PostgreSQL, Kafka, AWS EKS, Terraform
- Built distributed microservices for real-time CVE data processing using Kafka message streaming with schema validation. - Developed RAG (Retrieval-Augmented Generation) pipeline with Pinecone vector DB and Llama 3 LLM for natural language security queries and intelligent document retrieval. - Deployed on AWS EKS with Terraform; configured Prometheus/Grafana monitoring, alerting, and comprehensive observability.	
Multi-Tenant CRM Platform (Wasalt/Nesto)	React, TypeScript, Spring Boot, Node.js, PostgreSQL, Kafka, Redis
- Architected multi-tenant CRM with Spring Boot/Node microservices and IAM/RBAC supporting 50+ roles with tenant isolation. - Implemented Kafka streaming for notifications and gRPC for efficient inter-service communication; Redis achieving sub-100ms. - Built responsive React/TypeScript UI with real-time WebSocket updates for property listings and vendor dashboards.	
Open Source Contributions	GitHub: NeoSOFT-Technologies
Micro-Frontend Boilerplate: TypeScript + React + Webpack MF with Jest/Cypress testing; adopted by 15+ teams. Multi-Tenancy Admin Console: SaaS platform with RBAC, Keycloak SSO; Docker containerized with full CI/CD pipeline.	

Education & Certifications

Master of Science in Software Engineering Systems — Northeastern University, Boston, MA	Sept 2023 - May 2025
Relevant Coursework: Advanced Cloud Computing, Algorithms, Object-Oriented Design, Big Data Systems, Distributed Systems	
Bachelor of Engineering in Computer Engineering — University of Mumbai, Mumbai, India	Aug 2018 - Jun 2021
Certifications : Certified Java Enterprise Applications Developer (Oracle) OOP Using C++ (NIIT) Foundational AI (SkillUp)	